

Total no. of Pages: 2

Roll no.

Degree: BTech (CSMC)

Semester: III

END-SEMESTER THEORY EXAMINATION, NOV-DEC. 2024

Course Code: CMECC304

Course Title: Digital Logic Design

Time: 03:00 Hours

Max. Marks 50

Note: - Attempt all the five questions. Missing data/ information (if any), maybe suitably assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	A combinational circuit with three inputs, x, y, and z, and three outputs, A, B, and C. When the binary input is 0, 1, 2, or 3, the binary output is one greater than the input. When The binary input is 4, 5, 6, or 7, the binary output is two less than the input. (a) Draw the truth table. (b) Implement by using NOR gates only	5	CO1
1b	Express the function $Y=A+B'C$ in (a) Canonical POS (b) Canonical SOP	5	CO1
1c	Find the minimal SOP for the following Boolean expression using Quine McClusky method $F = \Sigma (1, 2, 3, 7, 8, 9, 11, 14, 15) + \text{Don't cares } (0, 10)$	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Realise a 32X1 multiplexer using two 8X1 multiplexers and suitable gates.	5	CO2
2b	Design a combinational circuit that converts a 4 bit Gray code to a BCD code.	5	CO2
2c	Design a 2 bit magnitude comparator and implement it by using a suitable size Decoder and OR/NOR gates.	5	CO2
Q 3	Attempt any 2 parts of the following		
3a	Design a synchronous mod 6 counter using JK flip flops.	5	CO3
3b	Differentiate between a latch and a flip flop. Realise a JK flip flop using an D flip flop.	5	CO3

3c	Distinguish between a Mealy and Moore machine using a state diagram. Convert the following Mealy machines to a Moore machine <table><tr><th>Presentation State</th><th>NS/Output X=0</th><th>NS/Output X=1</th></tr><tr><td>A</td><td>B,0</td><td>A,1</td></tr><tr><td>B</td><td>C,1</td><td>D,0</td></tr><tr><td>C</td><td>C, 0</td><td>E,1</td></tr><tr><td>D</td><td>A,0</td><td>B,0</td></tr><tr><td>E</td><td>D,1</td><td>C,1</td></tr></table>	Presentation State	NS/Output X=0	NS/Output X=1	A	B,0	A,1	B	C,1	D,0	C	C, 0	E,1	D	A,0	B,0	E	D,1	C,1	5	CO3
Presentation State	NS/Output X=0	NS/Output X=1																			
A	B,0	A,1																			
B	C,1	D,0																			
C	C, 0	E,1																			
D	A,0	B,0																			
E	D,1	C,1																			
Q 4	Attempt any 2 parts of the following																				
4a	Write the VHDL codes for a 8X1 multiplexer.	5	CO4																		
4b	Write the VHDL code for a 4 bit UP/DOWN counter using T flip-flop.	5	CO4																		
4c	Write short notes on the Modelling styles in VHDL.	5	CO4																		
Q 5	Attempt any 2 parts of the following																				
5a	Explain in detail Analog to digital converter. Also explain one type of ADC.	5	CO5																		
5b	Using a CMOS transistor, design a two input NAND gate and explain its working.	5	CO1																		
5c	Differentiate between PLA and ROM.	5	CO5																		